TNK1 AI Pipeline: Complete Model & Step-by-Step Technical Manual

Uncertainty-Aware Graph Neural Networks, Kinome Transfer Learning & Reinforcement Learning

Target: Tyrosine Kinase Non-Receptor 1 (TNK1, UniProt Q13470) | Architecture: Team A (AI) Closed-Loop Specification

Executive Problem Statement & Architectural Philosophy: Designing small-molecule inhibitors for Tyrosine Kinase Non-Receptor 1 (TNK1) faces an extreme low-data bottleneck: public databases contain fewer than 100 validated dose-response points for TNK1, whereas standard deep learning requires thousands of structures. This pipeline overcomes data scarcity by combining four tightly coupled mechanisms: (1) multi-stage transfer learning from the broader human kinome conditioned on protein binding-pocket embeddings, (2) calibrated epistemic and aleatoric uncertainty estimation so the system knows when it is guessing, (3) reinforcement learning driven by a pessimistic Lower Confidence Bound (LCB) reward to prevent generator exploitation, and (4) batch active learning coordinated every 3–4 weeks with Team B (Medicinal Chemistry).

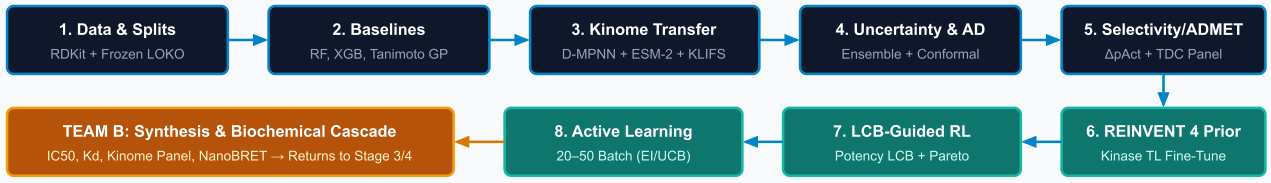


Figure 1: End-to-end closed-loop workflow linking Team A predictive/generative stages (1–8) with Team B experimental validation.

Stage 0: Environment, Reproducibility & Experiment Tracking

WEEKS 1–2

Before touching bioactivity data, Team A establishes deterministic execution and isolated environments. Because tiny datasets amplify stochastic variance, an unseeded run or silent package upgrade can masquerade as an algorithmic improvement.

- **Step 0.1** **Dual Conda Environment Isolation:** Create two strictly separated environments. Environment 1 (`tnk1-predict`, Python 3.10) houses PyTorch 2.x (compiled against CUDA 11.8/12.1), PyTorch Geometric (`torch_geometric`), RDKit (`2023.09+`), Chemprop v2, `scikit-learn`, `xgboost`, `gpytorch` (for custom Tanimoto Gaussian Processes), `fair-esm` (for ESM-2 protein embeddings), `mapie` (for conformal calibration), and PyTDC. Environment 2 (`tnk1-reinvent`) houses REINVENT 4 and its exact pinned dependencies. The two environments communicate during Reinforcement Learning (Stage 7) via a lightweight JSON/REST scoring server or CLI wrapper so package conflicts never occur.
- **Step 0.2** **Deterministic Seeding Protocol:** Every training, splitting, and sampling script must invoke a global seed lock across five random number generators: Python's built-in `random.seed(s)`, NumPy's `np.random.seed(s)`, PyTorch CPU `torch.manual_seed(s)`, PyTorch CUDA `torch.cuda.manual_seed_all(s)`, and CuDNN flags (`cudnn.deterministic = True`, `cudnn.benchmark = False`). Every model configuration is executed across **5 fixed random seeds** (`seeds = [42, 101, 202, 303, 404]`), and every table reports $\mu \pm \sigma$ across seeds.
- **Step 0.3** **Data Hash Locking & Experiment Tracking:** Every dataset generated in Stage 1 is saved as an immutable Parquet file alongside its SHA-256 checksum. Every training run logs its Git commit hash, dataset SHA-256 hash, exact hyperparameter YAML, and validation curves to Weights & Biases (W&B) or MLflow.

Stage 1: Multi-Source Data Engineering, Standardization & Splits WEEKS 1–4

Stage 1 constructs the foundational kinome bioactivity dataset and permanently freezes the evaluation splits. Every downstream model (Stages 2–5) reads from these exact files.

Step 1.1: Multi-Source Bioactivity Extraction

We extract all human kinase bioactivity records (roughly 500+ kinases) alongside every available record for **TNK1 (UniProt Q13470)** and its closest ACK-family relative **TNK2 / ACK1 (UniProt Q07912)**, plus non-receptor tyrosine kinase families (SRC, ABL1, LCK, FAK/PTK2, JAK, EGFR):

- **ChEMBL (v33/v34 SQLite dump):** Query the `activities`, `assays`, `target_dictionary`, and `compound_structures` tables. Filter for `target_type = 'SINGLE PROTEIN'`, `organism = 'Homo sapiens'`, `assay_type = 'B'` (biochemical binding/functional), and `standard_type IN ('IC50', 'Ki', 'Kd')`. Exclude records where `data_validity_comment IS NOT NULL`.
- **BindingDB & PubChem BioAssay:** Download the BindingDB flat TSV file and filter by UniProt IDs matching the human kinome list. Extract dose-response affinities; store single-dose percentage-inhibition screens separately for auxiliary pretraining.
- **PKIS, PKIS2 & Literature Kinome Scans:** Incorporate the Published Kinase Inhibitor Set (PKIS/PKIS2) matrices and Davis/Fabian/Karaman KINOMEscan K_d panels. These dense compound-by-kinase matrices are vital for learning cross-kinase selectivity.
- **Dark Kinase Knowledgebase (DKK) & Literature Curation:** Manually verify known multi-kinase binders with reported TNK1 affinity, including **TP-0903**, **dasatinib**, and **bosutinib**, ensuring their exact publication years and assay formats are recorded.

Step 1.2: RDKit Chemical Standardization Pipeline

Raw SMILES strings across databases represent identical molecules with inconsistent salt forms, protonation states, or tautomeric representations. If unstandardized, the same compound appears in both training and test sets (data leakage). Every SMILES passes through a 6-step RDKit standardization pipeline:

1. **Sanitization & Parsing:** Parse via `Chem.MolFromSmiles(smi, sanitize=True)`; drop invalid valence structures.
2. **Fragment & Salt Stripping:** Apply `rdMolStandardize.LargestFragmentChooser()` to strip counter-ions (e.g., HCl, mesylate, trifluoroacetate) and waters of crystallization, keeping the core organic pharmacophore.
3. **Functional Group Normalization & Reionization:** Apply `rdMolStandardize.Normalizer()` to fix nitro, sulfoxide, and azide representations, followed by `Reionizer()` and `Uncharger()` to bring molecules to a neutral canonical protonation state.
4. **Tautomer Canonicalization:** Apply `rdMolStandardize.TautomerEnumerator().Canonicalize(mol)` so keto-enol or 2-aminopyrimidine tautomers map to a single graph.
5. **Organic Medicinal Chemistry Filter:** Retain molecules with Molecular Weight $150 \le MW \le 900$ Da, heavy atom count $10 \le HAC \le 65$, and exclusively allowed elements $\{C, N, O, S, F, Cl, Br, I, P, B, H\}$ (discarding organometallics and inorganic complexes).
6. **Identity Key Generation:** Generate canonical isomeric SMILES and compute the 27-character `InChIKey`. The first 14 characters (the connectivity layer, `InChIKey_block1`) uniquely identify the stereoisomer-agnostic molecular skeleton and are used to prevent stereoisomer leakage across splits.

Step 1.3: Activity Transformation, Deduplication & Censored Labels

All affinities (IC_{50}, K_i, K_d) are converted to nanomolar (nM) and transformed into logarithmic potency: $pActivity = 9 - log_{10}(Value_{nM})$ (where 1 μM = 6.0, 100 nM = 7.0, 10 nM = 8.0). Because IC_{50} depends on ATP concentration via the Cheng-Prusoff equation whereas K_d and K_i are thermodynamic constants, `assay_type` is preserved as an explicit categorical feature. When multiple measurements exist for the same `(InChIKey_block1, uniprot_id, assay_type)` triplet:

- Compute the median $pActivity$ and standard deviation σ_{rep} across replicates. If the total range exceeds 1.0

Stage 2: Single-Task Baseline Models (Detailed Architectures) WEEKS 3–5

A complex deep-learning pipeline is only justified if it outperforms well-tuned classical machine learning on the exact same splits. In Stage 2, we implement five distinct baseline models trained exclusively on the TNK1 dataset (using the 5×5 nested scaffold cross-validation folds from Stage 1) and evaluate them on the held-out TNK1 test set.

Baseline Model	Input Representation	Key Hyperparameters & Tuning Grid	Primary Role in Benchmark
Model 2.1: k-NN Tanimoto	2048-bit Morgan FP (radius 2)	$k \in \{1, 3, 5, 7\}$, similarity-weighted mean	Sanity floor; tests pure structural memorization.
Model 2.2: Random Forest (RF)	2048-bit Morgan Count FP + 200 RDKit 2D descriptors	`n_estimators=500`, `max_features` ∈ `{'sqrt', 0.3, 0.5}`, `min_samples_leaf` ∈ `{1, 3, 5}`	Gold-standard tabular QSAR baseline; ensemble variance serves as baseline uncertainty.
Model 2.3: XGBoost Regressor	Same 2248-dim vector as RF (standardized)	`max_depth` ∈ `{3, 5, 7}`, `lr` ∈ `{0.01, 0.05}`, `subsample=0.8`, `colsample_bytree=0.7`, L2 `reg_lambda` ∈ `{1, 10}`	Tests gradient-boosted non-linear feature interactions on physicochemical properties.
Model 2.4: Exact Tanimoto GP	2048-bit binary/count Morgan FP	Tanimoto kernel K_{Tan}, trainable output scale α and Gaussian noise variance σ_n^2 via marginal log-likelihood	Strongest small-data Bayesian baseline; provides exact analytical posterior variance.
Model 2.5: Scratch D-MPNN	2D Molecular Graph $G = (V, E)$	3 message-passing steps, hidden dim 300, 2-layer FFN, dropout 0.2, AdamW `lr=1e-3`	Direct ablation proof: measures GNN performance *without* kinome transfer learning.

Step-by-Step Implementation of Each Baseline Model

Step 2.1 **Molecular Feature Engineering (Fingerprints + RDKit 2D Descriptors):** For Models 2.1–2.4, each molecule is converted into a 2048-dimensional Extended-Connectivity Fingerprint (ECFP4 / Morgan radius $r = 2$) using `AllChem.GetHashedMorganFingerprint(mol, radius=2, nBits=2048)` so substructure counts are preserved alongside binary presence. For RF and XGBoost, we concatenate 200 continuous 2D physicochemical descriptors from `rdkit.Chem.Descriptors` (including $MolLogP$, $TPSA$, $MolWt$, $NumHDonors$, $NumHAcceptors$, BalabanJ, BertzCT, and partial Gasteiger charges). Missing or infinite descriptor values are imputed using the training fold median and scaled via `RobustScaler`.

Step 2.2 **Model 2.1 (k-Nearest Neighbors Similarity Baseline):** Given a test molecule x, compute its Tanimoto similarity $S(x, x_i)$ to all training molecules x_i. Predict activity as the similarity-weighted average of the top k neighbors: $\hat{y} = \sum_{i=1..k} S(x, x_i)\, y_i \,/\, \sum_{i=1..k} S(x, x_i)$. If complex models fail to beat k-NN on scaffold splits, they are overfitting to scaffold-specific artifacts.

Step 2.3 **Models 2.2 & 2.3 (Random Forest & XGBoost):** Train `RandomForestRegressor(n_estimators=500)` and `XGBRegressor` inside the inner loop of the 5×5 nested scaffold CV. Append a one-hot encoded vector for `assay_type` (IC_{50}, K_i, K_d) so the trees can adjust for systematic offsets between binding and biochemical assays. For Random Forest, record the standard deviation across the 500 individual decision trees as an out-of-the-box uncertainty estimate.

Step 2.4 **Model 2.4 (Gaussian Process with Tanimoto Kernel):** In regimes under 100 data points, Gaussian Processes (GPs) are exceptionally strong because they do not overfit millions of weights. Standard RBF kernels perform poorly on binary/count fingerprints; instead, we implement the **Tanimoto (Jaccard) Kernel** in `GPyTorch`: $k_{Tan}(x, x') = \sigma_f^2 \cdot (\langle x, x' \rangle) \,/\, (\|x\|^2 + \|x'\|^2 - \langle x, x' \rangle)$. Given training matrix X and target vector y, the GP provides both the exact posterior mean $\mu(x_*) = k_*^T (K + \sigma_n^2 I)^{-1} y$ and exact epistemic variance $\sigma^2(x_*) = k(x_*, x_*) - k_*^T (K + \sigma_n^2 I)^{-1} k_*$. This GP serves as the benchmark for both Stage 3 (potency accuracy) and Stage 4 (uncertainty calibration).

Stage 3: Four-Stage Transfer Learning Architecture (Part A)

WEEKS 4–9

Stage 3 is the core algorithmic engine solving TNK1's data scarcity. Instead of learning medicinal chemistry from 60 compounds, the network learns general valence rules from 1 million compounds, learns ATP-pocket recognition from 300,000+ kinome measurements, focuses on ACK/SRC/ABL relatives, and finally adapts to TNK1.

3.1 Core Molecular Encoder: Directed Message-Passing Neural Network (D-MPNN)

Unlike standard Graph Convolutional Networks (GCNs) that associate messages with vertices (atoms) and suffer from message tottering (a message passing from atom v to w and immediately bouncing back to v), our primary encoder is a **Directed Message-Passing Neural Network (D-MPNN / Chemprop v2)** that propagates messages across *directed bonds (edges)* $v \rightarrow w$:

- **Atom Feature Vector $x_v \in \mathbb{R}^{133}$:** One-hot encoding of atomic number (H through I), total degree (0–5), formal charge (−2 to +2), chiral tag (CW, CCW, unspecified), number of bonded hydrogens (0–4), hybridization ($sp, sp^2, sp^3, sp^3d, sp^3d^2$), aromaticity flag (boolean), and scaled atomic mass.
- **Bond Feature Vector $e_{vw} \in \mathbb{R}^{14}$:** Bond type (single, double, triple, aromatic), conjugation flag, ring membership flag, and stereochemistry ($E, Z, cis, trans$).
- **Directed Edge Initialization & Message Passing ($t = 1 \dots T$, with $T = 3$ to 5 steps):** Initialize directed hidden states $h_{vw}^{0} = ReLU(W_i [x_v \,\|\, e_{vw}]) \in \mathbb{R}^{300}$. At each step t, aggregate incoming bonds to v *excluding* the reverse bond $w \rightarrow v$: $m_{vw}^{t+1} = \sum_{k \in N(v) \setminus \{w\}} h_{kv}^{t}$, and update $h_{vw}^{t+1} = ReLU(h_{vw}^{0} + W_m\, m_{vw}^{t+1})$.
- **Atom & Graph Readout:** After T steps, compute final atom representations $h_v = ReLU(W_a [x_v \,\|\, \sum_{k \in N(v)} h_{kv}^{T}])$ and pool across all atoms in molecule G using sum-plus-attention pooling to yield the molecular latent vector $z_{mol} \in \mathbb{R}^{300}$. Concatenate 200 normalized RDKit 2D descriptors to form $z_{mol+desc} \in \mathbb{R}^{500}$.



Figure 2: Target-conditioned D-MPNN architecture (Design B) combining 2D graph message passing with ESM-2 and KLIFS pocket representations.

3.2 Sub-Stage 1: Self-Supervised Molecular Pretraining (ChEMBL 1.5M)

Before introducing kinase labels, initialize the D-MPNN weights using self-supervised representation learning across ~1.5 million unlabelled drug-like compounds from ChEMBL (or load pretrained MolCLR / GROVER / Chemprop foundation weights). We combine two pretext objectives: (1) **Masked Atom/Substructure Prediction**, where 15% of graph nodes and their incident edges are masked and the encoder predicts their atomic number, degree, and aromaticity via cross-entropy loss; and (2) **Physicochemical Property Pretraining**, predicting 200 exact RDKit descriptors simultaneously. This ensures the initial message-passing filters recognize functional groups, ring systems, and hydrogen-bond donors/acceptors before seeing any protein target.

3.3 Sub-Stage 2: Kinome-Wide Multi-Task Training (Design A vs. Design B)

We implement and benchmark two distinct multi-task architectures across the ~300,000 compound–kinase pairs curated in Stage 1:

- **Design A (Multi-Head Shared Encoder):** The D-MPNN produces $z_{mol} \in \mathbb{R}^{500}$, which feeds into a linear layer with K parallel outputs ($\hat{y} \in \mathbb{R}^{K}$, one head per kinase). Because any given compound is only tested against a fraction of kinases, training uses a **Masked Huber Loss**: $L = \sum_{k=1..K} m_{ik} \cdot L_{Huber}(y_{ik}, \hat{y}_{ik})$ where mask $m_{ik} = 1$ if measured and 0 if missing. *Limitation:* TNK1's output head only receives gradients from the ~60 TNK1 data

Stage 3: Related-Kinase Weighting, Fine-Tuning & Ablations (Part B)

WEEKS 4–9

3.4 Sub-Stage 3: Related-Kinase Similarity Weighting & Censored Loss

Training equally on all 500 human kinases risks **negative transfer**: atypical kinases or lipid-like Ser/Thr kinases with tens of thousands of data points (e.g., PI3K, mTOR) can pull the shared latent space away from non-receptor tyrosine kinases. To focus the model on TNK1-relevant binding pockets:

- **Step 3.4.1** **Pocket & Domain Similarity Score $S(k, TNK1)$:** For every kinase k, compute the sequence identity over the 85 KLIFS pocket residues $Sim_{KLIFS}(k, TNK1) \in [0, 1]$ and the cosine similarity of the ESM-2 catalytic domain vectors $Sim_{ESM}(k, TNK1) \in [0, 1]$. Define the composite similarity $S_k = 0.7 \cdot Sim_{KLIFS} + 0.3 \cdot Sim_{ESM}$. Under this metric, **TNK2 (ACK1)** scores highest, followed by **SRC, ABL1, LCK, YES1, PTK2 (FAK), and JAK** families.
- **Step 3.4.2** **Sample Importance Weighting & Majority Capping:** Assign each training sample (G_i, k) an importance weight $w_{ik} = (1 + \lambda \cdot S_k^2) \cdot min(1.0, \sqrt{(2000 / N_k)})$, where N_k is the total compound count for kinase k (preventing EGFR or ChEMBL mega-targets with >15,000 compounds from dominating batches) and $\lambda \in \{0, 1, 3, 10\}$ is tuned on Leave-One-Kinase-Out (LOKO) validation.
- **Step 3.4.3** **Censored Asymmetric Loss Integration:** For exact measurements ($c_{ik} = 0$), apply Huber loss ($\delta = 1.0$). For right-censored weak negatives ($c_{ik} = 1$, e.g., $IC_{50} > 10\ \mu M \Rightarrow y_{ik} < \tau = 5.0$), penalize the model only if its prediction exceeds the threshold: $L_{cens} = max(0, \hat{y}_{ik} - \tau)^2$.

3.5 Sub-Stage 4: Few-Shot TNK1 Fine-Tuning with L2-SP Regularization

Once the kinome-conditioned checkpoint θ_0 is trained, we adapt it to the TNK1 training split without catastrophic forgetting using a 3-step protocol:

1. **Phase 1 — Linear Probing (Epochs 1–15):** Freeze the D-MPNN graph encoder (θ_{enc}) and the kinase projection layer (θ_{kproj}). Train only the final fusion MLP head (θ_{head}) on TNK1 data using AdamW with learning rate $\eta = 3 \times 10^{-4}$. This aligns the output scale to the specific TNK1 assay distribution.
2. **Phase 2 — Discriminative Unfreezing (Epochs 16–60):** Unfreeze the final message-passing layer and readout layers using discriminative learning rates ($\eta_{enc} = 1 \times 10^{-5}$, $\eta_{head} = 1 \times 10^{-4}$) with cosine annealing and early stopping on the inner CV validation fold.
3. **Phase 3 — L2-Starting-Point (L2-SP) Elastic Anchor:** Standard L2 weight decay pulls weights toward zero, destroying pretrained features. Instead, we apply **L2-SP regularization**, which penalizes deviation from the pretrained kinome checkpoint θ_0 : $L_{total}(\theta) = L_{task}(\theta) + \alpha_{enc} \|\theta_{enc} - \theta_{0,enc}\|_2^2 + \alpha_{head} \|\theta_{head} - \theta_{0,head}\|_2^2$, with $\alpha_{enc} = 10^{-2}$ and $\alpha_{head} = 10^{-3}$.

```
# Stage 3.5: PyTorch Implementation of L2-SP Fine-Tuning Loss + Censored Activity
Handling def compute_finetune_loss(pred_mu, target_y, is_censored, model,
pretrained_state_dict, alpha_sp=1e-2): exact_mask = (is_censored == 0) cens_mask =
(is_censored == 1) # e.g., IC50 > 10 uM means true pActivity < target_y (5.0) loss_exact =
F.huber_loss(pred_mu[exact_mask], target_y[exact_mask], delta=1.0) if exact_mask.any() else
0.0 loss_cens = torch.mean(F.relu(pred_mu[cens_mask] - target_y[cens_mask]) ** 2) if
cens_mask.any() else 0.0 l2_sp_reg = sum(torch.sum((p -
pretrained_state_dict[n].to(p.device)) ** 2) for n, p in model.named_parameters() if
p.requires_grad and n in pretrained_state_dict) return loss_exact + 0.5 * loss_cens +
alpha_sp * l2_sp_reg
```

3.6 Mandatory Stage 3 Ablation Matrix (Deliverable Benchmark)

Ablation ID & Model Configuration	Pretraining Source	Kinase Conditioning	TNK1 Fine-Tuning	What This Row Proves in the Paper

Stage 4: Uncertainty Quantification, Calibration & Applicability Domain

WEEKS 6–11

When a generative model proposes novel structures for a target with <100 known ligands, point predictions are dangerous: the generator will inevitably discover regions of chemical space where the neural network outputs artificially high potency due to uncalibrated extrapolation. Stage 4 equips the Stage 3 predictor with decomposed uncertainty and statistical coverage guarantees.

4.1 Four Uncertainty Architectures & Exact Variance Decomposition

We distinguish **aleatoric uncertainty (σ_{ale}^2)**—irreducible experimental noise inherent to biochemical IC_{50} / K_d assays—from **epistemic uncertainty (σ_{epi}^2)**—reducible model ignorance caused by lack of training molecules near a query scaffold:

- **Architecture 4.1 (Primary Production Model — Deep Ensemble of Heteroscedastic D-MPNNs):** Train $M = 10$ independent instances of the Stage 3.5 model (A7) using different random seeds ($m = 1 \dots 10$), different bootstrap/CV data orderings, and independent weight initializations. Each ensemble member m outputs two scalars per molecule–kinase pair: predicted mean $\mu_m(x)$ and predicted log-aleatoric variance $s_m(x) = \log \sigma_{ale,m}^2(x)$, trained via the **Gaussian Negative Log-Likelihood (NLL) Loss**: $L_{NLL} = \frac{1}{2} \exp(-s_m(x)) \cdot (y - \mu_m(x))^2 + \frac{1}{2} s_m(x)$. At inference, applying the law of total variance across the $M = 10$ members decomposes total variance cleanly:
  • Ensemble Mean Prediction: $\mu(x) = (1/M) \sum_{m=1..M} \mu_m(x)$
  • Aleatoric Variance (Assay Noise): $\sigma_{ale}^2(x) = (1/M) \sum_{m=1..M} \exp(s_m(x))$
  • Epistemic Variance (Model Ignorance): $\sigma_{epi}^2(x) = (1/M) \sum_{m=1..M} (\mu_m(x) - \mu(x))^2$
  • Total Predictive Standard Deviation: $\sigma_{tot}(x) = \sqrt{(\sigma_{ale}^2(x) + \sigma_{epi}^2(x))}$.
- **Architecture 4.2 (Monte Carlo Dropout):** Keep dropout layers ($p = 0.2$) active in both the D-MPNN readout and fusion MLP during inference and run $T = 50$ stochastic forward passes. Compute mean and variance across the 50 passes. Used as a lightweight single-checkpoint comparison.
- **Architecture 4.3 (Deep Evidential Regression):** Replace the final layer with a 4-output head parameterizing a Normal-Inverse-Gamma (NIG) conjugate prior $(\gamma, \nu, \alpha, \beta)$ with constraints $\nu > 0, \alpha > 1, \beta > 0$ enforced via Softplus. In a single forward pass, prediction is $\mathbb{E}[\mu] = \gamma$, aleatoric variance is $\mathbb{E}[\sigma^2] = \beta / (\alpha - 1)$, and epistemic variance is $Var(\mu) = \beta / (\nu(\alpha - 1))$.

Uncertainty Method	Forward Passes	Separates σ_{epi} / σ_{ale} ?	Calibration Quality on LOKO/Scaffold	Role in Pipeline
Heteroscedastic Deep Ensemble ($M=10$)	10 passes	Yes (Total Variance Law)	Highest (Gold standard)	Primary scorer for Stage 7 RL & Stage 8 AL.
Tanimoto Gaussian Process (Stage 2.4)	1 kernel pass	Yes (Analytical)	High on small data	Secondary audit scorer for reward-hacking check.
Monte Carlo Dropout ($T=50$)	50 passes	Approx. epistemic	Moderate (often overconfident)	Ablation baseline in Stage 4 report.
Evidential Regression (NIG Head)	1 pass	Yes (Single pass)	Sensitive to regularization coeff	Fast pre-filter for 100k+ virtual libraries.

4.2 Split Adaptive Conformal Prediction & Temperature Recalibration

Raw neural network standard deviations $\sigma_{tot}(x)$ are frequently miscalibrated (e.g., a nominal $\pm 1.645\sigma$ interval may cover only 72% of true test points instead of 90%). We apply two post-hoc calibration layers:

1. **Variance Temperature Scaling:** On the out-of-fold validation predictions, optimize a scalar multiplier $\tau_{temp} > 0$ minimizing Gaussian NLL: $\sigma_{cal}(x) = \tau_{temp} \cdot \sigma_{tot}(x)$.

Stage 5: Multi-Target Selectivity, ADMET & Medicinal Chemistry Filters

WEEKS 8–12

A nanomolar TNK1 binder that simultaneously inhibits SRC, ABL1, KDR, or hERG is a promiscuous toxin, not a drug lead. Stage 5 builds the auxiliary property predictors that constrain the generator to selective, orally bioavailable, synthetically tractable chemical space.

5.1 Ensemble-Propagated Kinome Selectivity Model

Define the critical off-target panel $\mathcal{O} = \{TNK2\ (ACK1), SRC, ABL1, LCK, PTK2\ (FAK), EGFR, KDR\ (VEGFR2), FLT3\}$ in consultation with Team B. Because the Stage 3–4 Heteroscedastic Deep Ensemble ($M = 10$) is conditioned on kinase embeddings, each member $m \in \{1 \dots 10\}$ predicts potency for TNK1 and every off-target $k \in \mathcal{O}$ simultaneously. To preserve covariance across related kinases, we compute selectivity *inside each ensemble member* m *first*, using both worst-case hard margin and soft LogSumExp margin:

- **Member-Level Selectivity Margin:** $\Delta S_m(x) = \mu_m(x, TNK1) - max_{k \in \mathcal{O}} \mu_m(x, k)$ (where $\Delta S = 1.5$ corresponds to a $10^{1.5} \approx 31.6$ -fold selectivity window, meeting Team B's 30-fold TPP target).
- **Selectivity Mean & Uncertainty:** $\mu_{sel}(x) = (1/M) \sum_{m=1..M} \Delta S_m(x)$ and $\sigma_{sel}(x) = \sqrt{((1/M) \sum_{m=1..M} (\Delta S_m(x) - \mu_{sel}(x))^2)}$. Sanity check: known pan-kinase inhibitors (dasatinib, bosutinib) must yield $\mu_{sel} \le 0$.

5.2 Multi-Task ADMET Predictor Suite (Therapeutics Data Commons)

We train a dedicated 5-member Chemprop v2 D-MPNN ensemble (augmented with RDKit 2D features, evaluated on scaffold splits) across seven core ADMET endpoints from Therapeutics Data Commons (TDC) / ADMET-AI:

ADMET Endpoint	TDC Training Dataset	Model Task & Output	Target Product Profile (TPP) Threshold
Aqueous Solubility	`AqSolDB` (9,982 cpds)	Regression ($log\ S$ in mol/L)	$log\ S \ge -5.0$ (≥ 10 μM kinetic solubility)
Intestinal Permeability	`Caco2_Wang` (906 cpds)	Regression ($log\ P_{app}$ cm/s)	$log\ P_{app} > -5.15$ (high permeability)
Microsomal Stability	`Clearance_Microsome_AZ`	Regression ($log\ CL_{int}$ mL/min/kg)	Low-to-moderate intrinsic clearance (< 30)
CYP3A4 / CYP2D6 Inhibition	`CYP3A4_Veith`, `CYP2D6_Veith`	Binary Classification (P_{inhib})	$P_{inhib} < 0.35$ (avoid DDI liabilities)
hERG Cardiotoxicity	`hERG_Karim` (13,445 cpds)	Binary Classification (P_{hERG} blocker)	$P_{hERG} < 0.20$ (strict safety penalty)
AMES Mutagenicity	`AMES` (7,278 cpds)	Binary Classification (P_{AMES})	$P_{AMES} < 0.15$ (hard rejection if ≥ 0.5)

5.3 Rule-Based Structural Filters, Hinge Pharmacophore & SA Score

Every candidate is evaluated with rapid deterministic RDKit filters: (1) **Quantitative Estimate of Drug-likeness (QED ∈ [0, 1])**; (2) **Ertl Synthetic Accessibility Score (SA ∈ [1, 10])**, targeting $SA \le 4.2$; (3) **MedChem Structural Alerts:** zero matches allowed across RDKit `FilterCatalog` for **PAINS (A, B, C)**, **Brenk reactive/toxicophoric groups** (michael acceptors, acyl halides, epoxides, nitroaromatics), and **NIH** filters; and (4) **Hinge-Binder SMARTS Verification:** optional soft bonus for containing validated ATP-hinge H-bond donor/acceptor motifs (e.g., 2-aminopyrimidine, pyrrolo[2,3-d]pyrimidine, quinazoline, indazole, or pyrazolopyrimidine).

Stage 6: Molecular Generator Architecture & Kinase Prior Tuning

WEEKS 8–12

Stage 6 builds the generative policy π_θ that proposes novel chemical structures before reinforcement learning begins.

Stage 7: Reinforcement Learning with Uncertainty-Aware LCB Reward

WEEKS 10–15

In standard RL drug design, the policy maximizes predicted potency $\mu(x)$. In low-data regimes like TNK1, this causes catastrophic **reward hacking (adversarial exploitation)**: the generator drifts into out-of-distribution chemical space where the neural network hallucinates $pIC_{50} > 9.0$ with huge epistemic variance σ_{epi}. Stage 7 eliminates this failure mode by optimizing the **Lower Confidence Bound (LCB)**.

7.1 Multi-Objective Reward Transformations & Geometric Aggregation

Every reward component $r_j(x)$ is mapped smoothly onto $[0, 1]$ using calibrated double-sigmoid or reverse-sigmoid transforms, and aggregated via a **Weighted Geometric Mean** gated by a hard medicinal chemistry filter $I_{hard}(x) \in \{0, 1\}$: $R(x) = I_{hard}(x) \cdot \prod_{j=1..7} [\, max(r_j(x), 10^{-4})\,]^{w_j / \sum w_k}$. Because it is a product, a molecule that scores 0.99 on potency but fails hERG or synthetic accessibility ($r_j \to 0$) receives a near-zero total reward.

Reward Component $r_j(x)$	Mathematical Formulation (Mapped to $[0, 1]$)	Weight w_j	Role in Preventing Failure Modes
1. TNK1 Potency LCB	$r_{pot} = \sigma_{sig}(\, \mu_{TNK1}(x) - \beta(t) \cdot \sigma_{epi,TNK1}(x);\ c=7.0,\ k=2.0\,)$	$w_1 = 3.0$	Rewards $IC_{50} < 100$ nM only when ensemble members agree (σ_{epi} small).
2. Selectivity LCB	$r_{sel} = \sigma_{sig}(\, \mu_{sel}(x) - \gamma \cdot \sigma_{sel}(x);\ c=1.2,\ k=2.0\,)$	$w_2 = 2.0$	Enforces ≥15–30× selectivity over ACK1, SRC, ABL1, FAK, EGFR, KDR.
3. Applicability Domain	$r_{AD} = exp(\, -max(0, d_{lat}(x) - d_{thr}) / 0.15\,)$	$w_3 = 1.5$	Smoothly penalizes latent/Tanimoto drift outside the Stage 4.3 AD boundary.
4. Synthetic Accessibility	$r_{SA} = clip(\, (5.5 - SA(x)) / (5.5 - 2.0),\ 0,\ 1\,)$	$w_4 = 1.0$	Forces simple, makeable rings ($SA \le 3.5$ ideal; >5.5 zeroed).
5. ADME Composite	Geometric mean of $r_{AqSol}, r_{Caco2}, r_{Microsome}$	$w_5 = 1.0$	Ensures oral absorption and metabolic stability.
6. Safety (hERG/CYP/AMES)	$(1 - P_{hERG}) \cdot (1 - P_{AMES}) \cdot (1 - 0.5\, P_{CYP3A4})$	$w_6 = 1.5$	Prevents cardiotoxic lipophilic amines and mutagens.
7. Drug-Likeness (QED)	$r_{QED} = QED(x) \in [0, 1]$	$w_7 = 0.5$	Maintains balanced MW, LogP, TPSA, and rotatable bonds.
Hard Gate $I_{hard}(x)$	0 if invalid, PAINS/Brenk match, $MW > 600$, or $P_{AMES} > 0.5$; else 1	Multiplier	Immediate zero reward for reactive or toxicophoric structures.

7.2 Policy Optimization Algorithm, $\beta(t)$ Schedule & Memory Bucket Filter

We optimize the agent network π_θ (initialized from the kinase prior P_{kinase}) over $T = 1{,}000$ steps (batch size $B = 128$ molecules/step) using REINVENT's **Augmented Log-Likelihood Loss**, which embeds a built-in Kullback-Leibler (KL) anchor to P_{kinase}: $L_{RL}(\theta) = (1/B) \sum_{i=1..B} [\, \log P_{kinase}(x_i) + \sigma_{RL} \cdot R(x_i) - \log \pi_\theta(x_i)\,]^2$, with $\sigma_{RL} = 120$ and Adam learning rate 10^{-4}.

- **Scheduled Exploration-to-Exploitation ($\beta(t)$ Annealing):** During steps 1–300, set $\beta = 0.3$ so the policy can explore diverse scaffolds even if uncertainty is moderate. From steps 301–1000, linearly anneal β from $0.3 \to 1.5$, forcing the policy to converge exclusively onto scaffolds with high ensemble confidence.
- **Bemis-Murcko Diversity Filter (Preventing Mode Collapse):** Maintain a scaffold memory table keyed by Murcko scaffold SMILES. Once any scaffold accumulates $N_{max} = 25$ molecules with $R(x) \ge 0.55$, any subsequent molecule sharing that scaffold receives $R(x) = 0$, compelling the agent to discover multiple distinct chemical series.

Stage 8: Closed-Loop Active Learning & Batch Acquisition Protocol

WEEKS 14–52 (3–4 WK CYCLES)

Stage 8 bridges Team A (AI) and Team B (Medicinal Chemistry). Every 3–4 weeks, the pipeline compresses 50,000+ generated candidates into a diverse, synthetically feasible batch of $N_{batch} = 40$ **compounds** for wet-lab synthesis and biochemical testing, then ingests the experimental results to retrain and recalibrate.

8.1 Step-by-Step Cycle Execution & Acquisition Math

1. **Step 8.1** **Candidate Pool Generation & Hard Filtering:** Pool 50,000 molecules sampled from the latest RL agents. Strip duplicates against all historical ChEMBL and in-house synthesized compounds, enforce $I_{hard}(x) = 1$, require cross-architecture audit agreement ($|\mu_{DMPNN} - \mu_{GP}| \le 1.0$), and filter to top 1,500 by Pareto rank.
2. **Step 8.2** **Computer-Aided Retrosynthesis Pre-Screen (AiZynthFinder) & Docking Pose Check:** Run Monte Carlo Tree Search retrosynthesis in `AiZynthFinder` against commercial building-block stock (Enamine / eMolecules / Sigma). Retain only compounds solvable in ≤ 4 synthetic steps with state score ≥ 0.85. In parallel, Team B docks the surviving ~400 molecules into the TNK1 homology/AlphaFold DFG-in/out pocket to verify hinge hydrogen bonding and absence of severe steric clashes.
3. **Step 8.3** **Acquisition Function Scoring (EI and UCB):** Let y^+ be the highest experimental pIC_{50} measured for TNK1 so far. Using the calibrated ensemble mean $\mu(x)$ and epistemic uncertainty $\sigma_{epi}(x)$, compute both **Expected Improvement (EI)** and **Upper Confidence Bound (UCB)**:
   • $EI(x) = (\mu(x) - y^+ - \xi)\,\Phi(Z) + \sigma_{epi}(x)\,\phi(Z)$, where $Z = (\mu(x) - y^+ - \xi) / \sigma_{epi}(x)$, $\xi = 0.05$, and Φ, ϕ are the standard normal CDF and PDF.
   • $UCB(x) = \mu(x) + \kappa \cdot \sigma_{epi}(x)$ with $\kappa = 1.5$.
4. **Step 8.4** **Butina Sphere-Exclusion Clustering & Stratified 40-Compound Batch Selection:** Cluster the retrosynthesis-verified pool using the Butina algorithm on Morgan fingerprints (Tanimoto distance cutoff $d = 0.40$, i.e., similarity ≤ 0.60 between cluster centroids) and select at most 2 analogs per cluster (1 centroid + 1 matched-pair analog for instant SAR) across four stratified buckets:

Batch Bucket	Allocation (N=40)	Selection Criterion (Within Butina Clusters)	Experimental Purpose in Closed Loop
1. Exploitation Slice	16 cpds (40%)	Maximize $LCB(x) = \mu(x) - 1.0 \cdot \sigma_{epi}(x)$ with $\mu_{sel} \ge 1.5$	Delivers immediate sub-100 nM selective hits and validates model confidence.
2. Exploration Slice	12 cpds (30%)	Maximize $EI(x)$ or $UCB(x)$ subject to $\mu(x) \ge 6.3$	Probes high-uncertainty pockets of chemical space; yields maximum information gain.
3. Novel Scaffold Slice	8 cpds (20%)	Top-scoring centroids of unrepresented Murcko scaffolds	Guarantees scaffold hopping and broad IP coverage.
4. Control Slice	4 cpds (10%)	Random/baseline-RF picks passing identical medchem filters	Provides prospective statistical control arm for the final paper.

8.2 Ingesting Wet-Lab Results, Upweighted Retraining & Recalibration

When Team B returns biochemical IC_{50} / K_d results (purity ≥ 95% by HPLC/NMR), Team A executes a strict **Evaluate-Before-Train Protocol**: (1) Score the returning 40 compounds using the *frozen* pre-batch ensemble to record prospective RMSE, Spearman ρ, hit rate ($IC_{50} < 1\,\mu M$ and $< 100\,nM$), and actual 90% conformal coverage; (2) Append all 40 results—including confirmed inactives ($> 10\,\mu M$ as censored labels)—into the TNK1 dataset with sample weight $w_{inhouse} = 3.0$ (upweighting consistent internal assay data over noisy literature data); and (3) Re-run Stage 3.5 L2-SP fine-tuning, update the 10-member ensemble, recompute the conformal quantile $\hat{q}_{0.90}$, and expand the Stage 4.3 Applicability Domain boundary to include the newly validated scaffolds.

Stage 9: Comprehensive Evaluation, Statistical Proofs & Governance

MONTHS 11–12

Master Model Blueprint, Team B Data Contract & Execution Checklist

REFERENCE BLUEPRINT

Pillar 4 (Headline Prospective Wet-Lab Claim): Across all DMTA cycles, report the cumulative experimental hit rate ($\% \; with \; IC_{50} < 1 \; \mu M$ and $< 100 \; nM$) with exact Clopper-Pearson 95% binomial confidence intervals for the Exploitation Slice vs. Exploration Slice vs. Control Slice, plus the potency trajectory from Cycle 1 to lead series ($IC_{50} < 100 \; nM$, $\geq 30\times$ selectivity).

Master Reference Table: Every Model Implemented Across Stages 1–8

Pipeline Stage	Exact Model / Algorithm Name	Input Features & Dimensionality	Loss / Objective Function	Primary Library / Tool
Stage 2 (Baselines)	k-NN, Random Forest (500 trees), XGBoost	2048-bit Morgan Count FP + 200 RDKit 2D (2248d)	Squared Error / Tree Variance	`scikit-learn`, `xgboost`
Stage 2.4 & 7.4	Exact Tanimoto Gaussian Process	2048-bit Morgan Count FP	Exact Marginal Log-Likelihood	`GPyTorch` (Custom Kernel)
Stage 3.1–3.2	Self-Supervised D-MPNN Encoder	Atom (133d) + Directed Bond (14d) Graph	15% Masked Atom CE + 200 RDKit MSE	`Chemprop v2`, `PyG`
Stage 3.3–3.5	Kinase-Conditioned D-MPNN (Design B)	z_{mol} (500d) + ESM-2 (1280d) + KLIFS (425d)	Similarity-Weighted Huber + Censored + L2-SP	`PyTorch`, `fair-esm`
Stage 4.1	10-Member Heteroscedastic Deep Ensemble	10 independent seeds of Design B (631d fused)	Heteroscedastic Gaussian NLL	`PyTorch` Ensemble Wrapper
Stage 4.2–4.3	Split Conformal + Dual-Space 5-NN AD	Normalized residuals $\|y-\mu\|/\sigma$; z_{mol} & Morgan FP	Empirical $(1-\alpha)$ Quantile + 95th %ile AD	`MAPIE`, `RDKit`
Stage 5.2	7-Endpoint Multi-Task ADMET Ensembles	2D Molecular Graph + 200 RDKit descriptors	MSE (Reg) / Binary Cross-Entropy (Class)	`Chemprop v2`, `PyTDC`
Stage 6 & 7	REINVENT 4 Autoregressive SMILES Policy	Randomized SMILES Token Sequences	Augmented Likelihood + LCB Geometric Reward	`REINVENT 4`
Stage 8	EI / UCB + Butina + AiZynthFinder MCTS	Calibrated $(\mu, \sigma_{epi}, \mu_{sel})$ + Morgan FP	Expected Improvement + Pareto Dominance	`RDKit Butina`, `AiZynthFinder`

Shared Data Contract (Team A ↔ Team B Interface Schema)

FROZEN WEEK 1

Column Name	Data Type	Populated By	Specification & Validation Rule
`compound_id` / `cycle_id`	`string` / `int`	Team A	Unique identifier `TNK1-C01-001` and DMTA cycle number (1, 2, 3...)
`canonical_smiles` / `inchikey`	`string`	Team A	RDKit standardized canonical SMILES and 27-char InChIKey
`batch_slice`	`enum`	Team A	`EXPLOIT`, `EXPLORE`, `DIVERSITY`, `CONTROL`, or `REF_STD`
`pred_mu_tnk1` / `pred_sigma_epi`	`float` / `float`	Team A	Pre-synthesis predicted $pActivity$ and epistemic σ_{epi} (frozen before assay)
`pred_ci90_low` / `pred_ci90_high`	`float` / `float`	Team A	Conformal 90% lower and upper bounds $[\mu - q\hat{}\sigma, \mu + q\hat{}\sigma]$